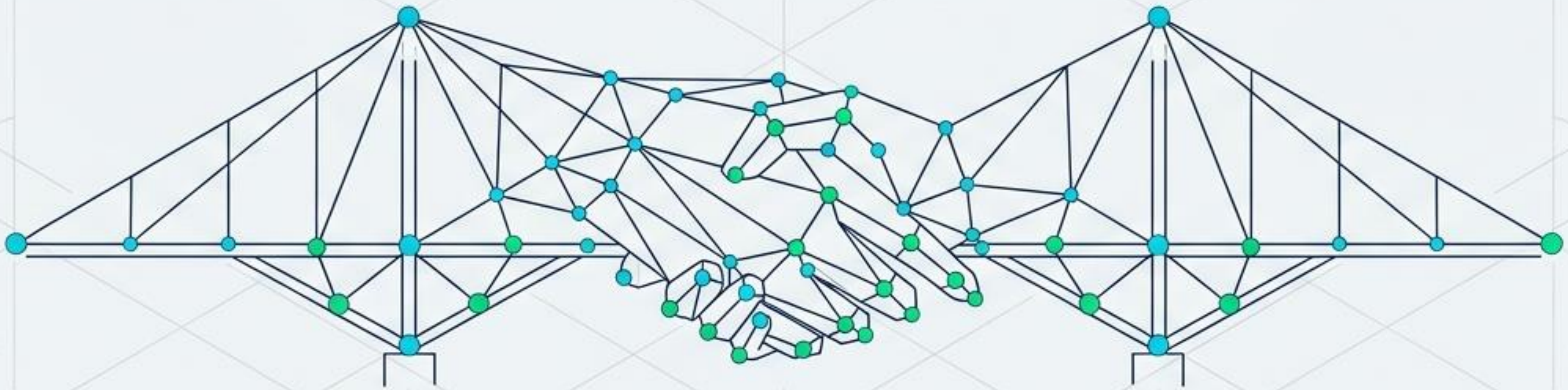
SIGN-BRIDGE

Bridging the Communication Chasm with Real-Time CNNs



Sabin Sasidharan

M.Sc. Computer Science

N.S.S. College Ottapalam

THE PROBLEM

430 Million

Individuals globally with disabling hearing loss who rely on sign language as their primary mother tongue.

The Communication Chasm: A massive daily barrier in banks, hospitals, hospitals, and schools requiring a technological translator.

<1%

Of the global hearing population possesses even basic sign language proficiency.

THE LITERATURE GAP

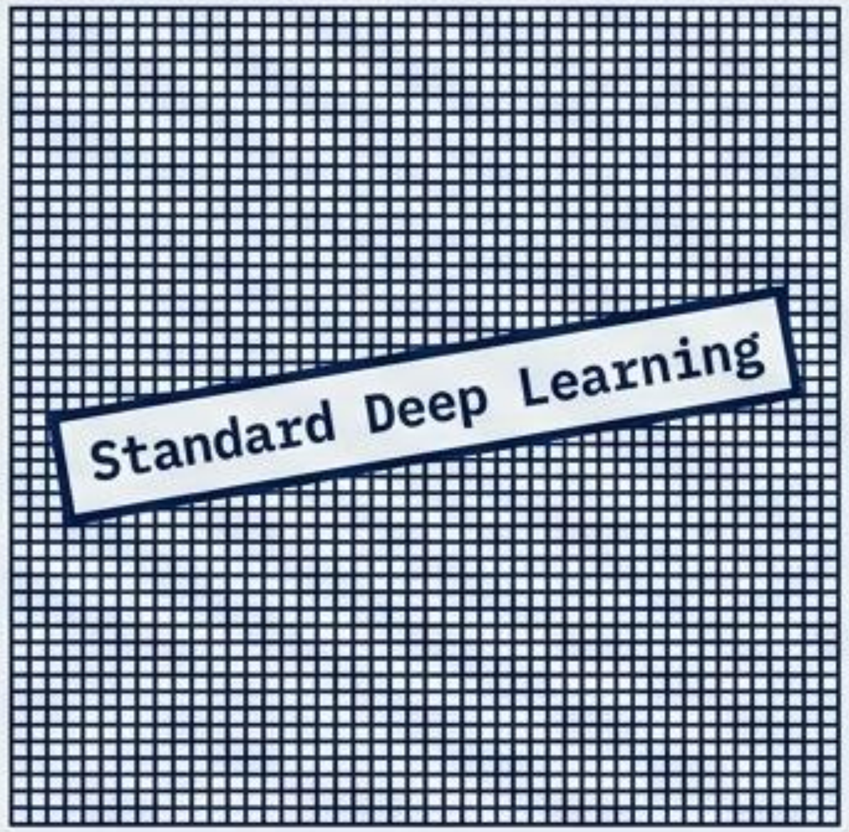
THE EVOLUTION OF SIGN RECOGNITION		
GENERATION 1: HARDWARE SENSORS	GENERATION 2: RAW PIXEL AI	GENERATION 3: SIGN-BRIDGE
COST: $5k - $15k (Gloves)	COST: Requires expensive GPUs	COST: $0 (Uses existing webcam)
HARDWARE: Heavy Wearables	HARDWARE: High-end compute only	HARDWARE: Basic Laptop CPU
LIGHTING: Invariant	LIGHTING: Drops 30% accuracy in bad light	LIGHTING: 100% Invariant
VERDICT: Too Expensive & Uncomfortable	VERDICT: Fails in real-world conditions	VERDICT: The Accessible Solution

By moving away from expensive hardware and heavy pixel grids, SIGN-BRIDGE optimizes for practical, living-room deployability.

THE SOLUTION: GEOMETRY OVER PIXELS

The 15,000-Fold Breakthrough

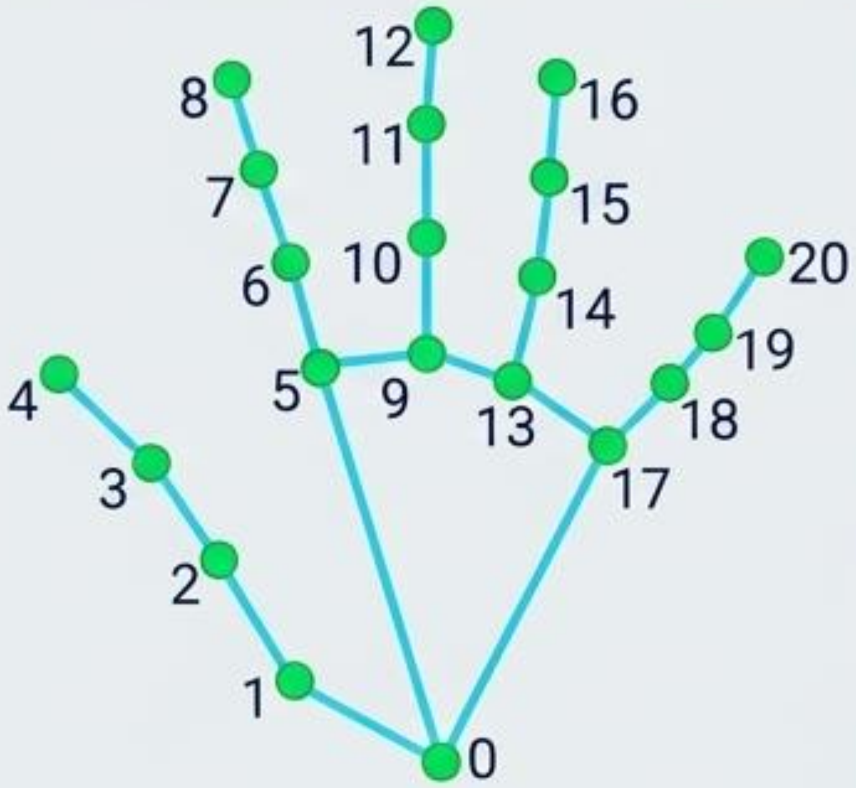
Raw Image Processing



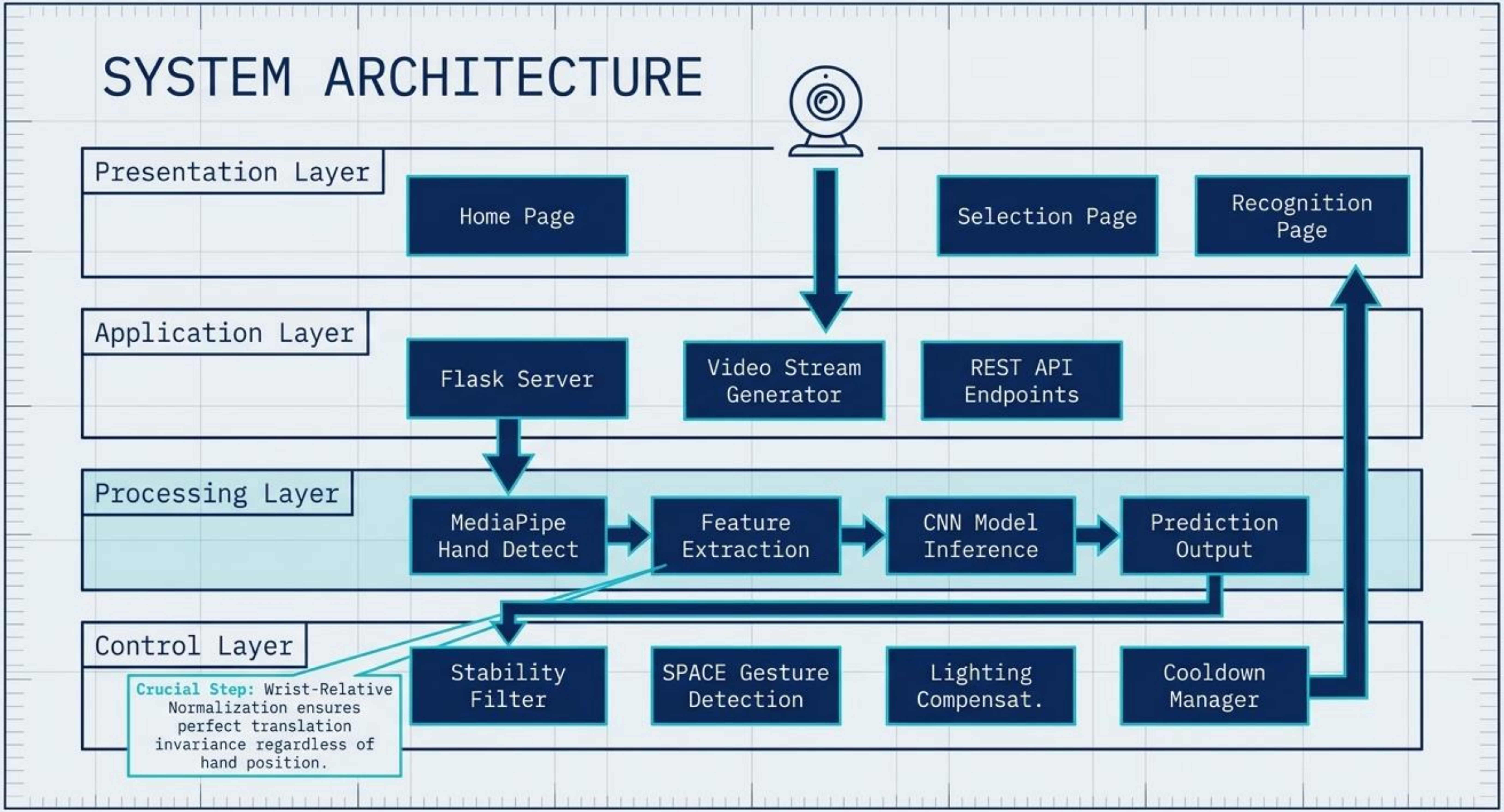
Processes heavy image files with thousands of pixels, requiring massive GPU power.

15,000x Data Reduction = Sub-5ms Processing on Basic CPUs

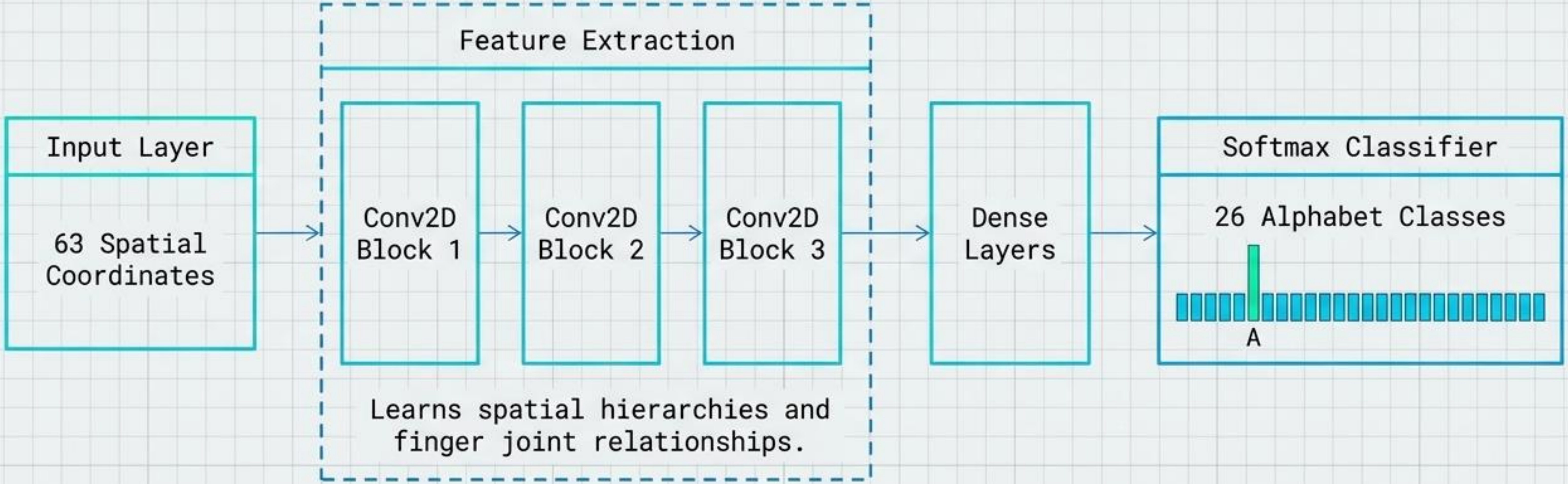
SIGN-BRIDGE Approach



Extracts a lightweight blueprint of 21 3D coordinates (63 total numbers per hand) via MediaPipe.


SYSTEM ARCHITECTURE
Presentation Layer
Home Page
Selection Page
Recognition Page
Application Layer
Flask Server
Video Stream Generator
REST API Endpoints
Processing Layer
MediaPipe Hand Detect
Feature Extraction
CNN Model Inference
Prediction Output
Control Layer
Stability Filter
SPACE Gesture Detection
Lighting Compensat.
Cooldown Manager
Crucial Step: Wrist-Relative Normalization ensures perfect translation invariance regardless of hand position.

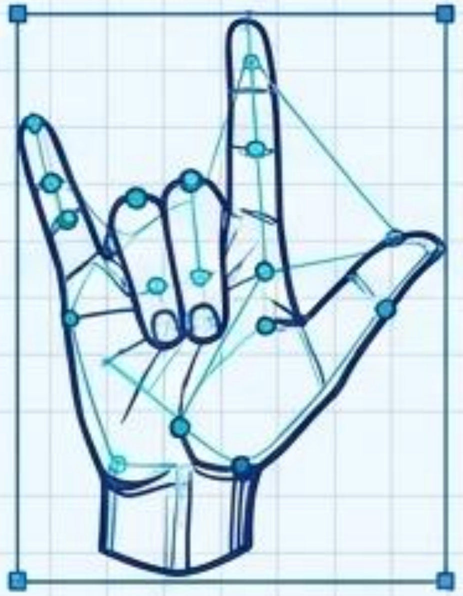
THE BRAIN: LIGHTWEIGHT CNN



Total Architecture: Only 210,000 Parameters. Tiny, blisteringly fast, and completely removes the need for manual feature engineering.

MULTI-MODAL DESIGN: ASL vs. ISL

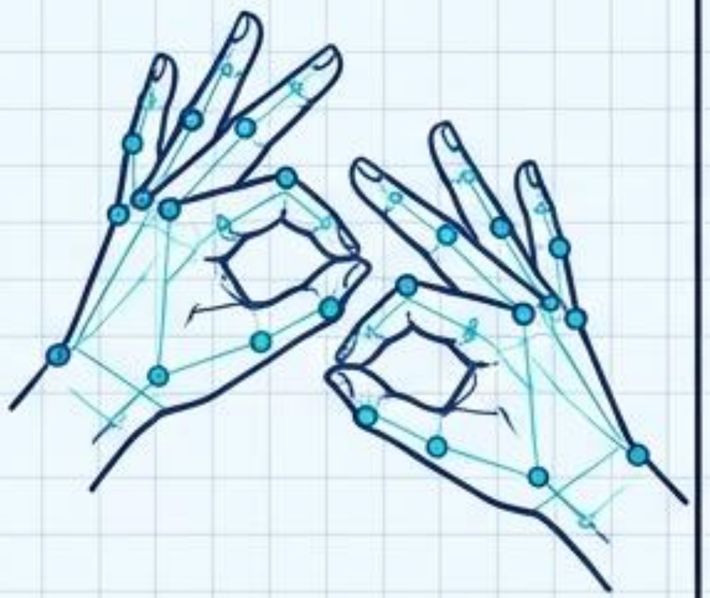
American Sign Language (ASL)



Input:
Single-Handed

Dimensionality:
63 Features

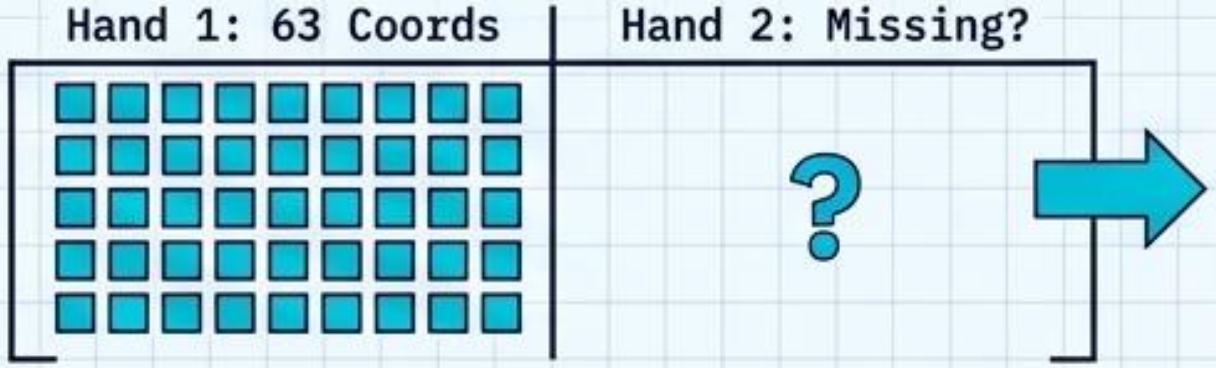
Indian Sign Language (ISL)



Input:
Dual-Handed

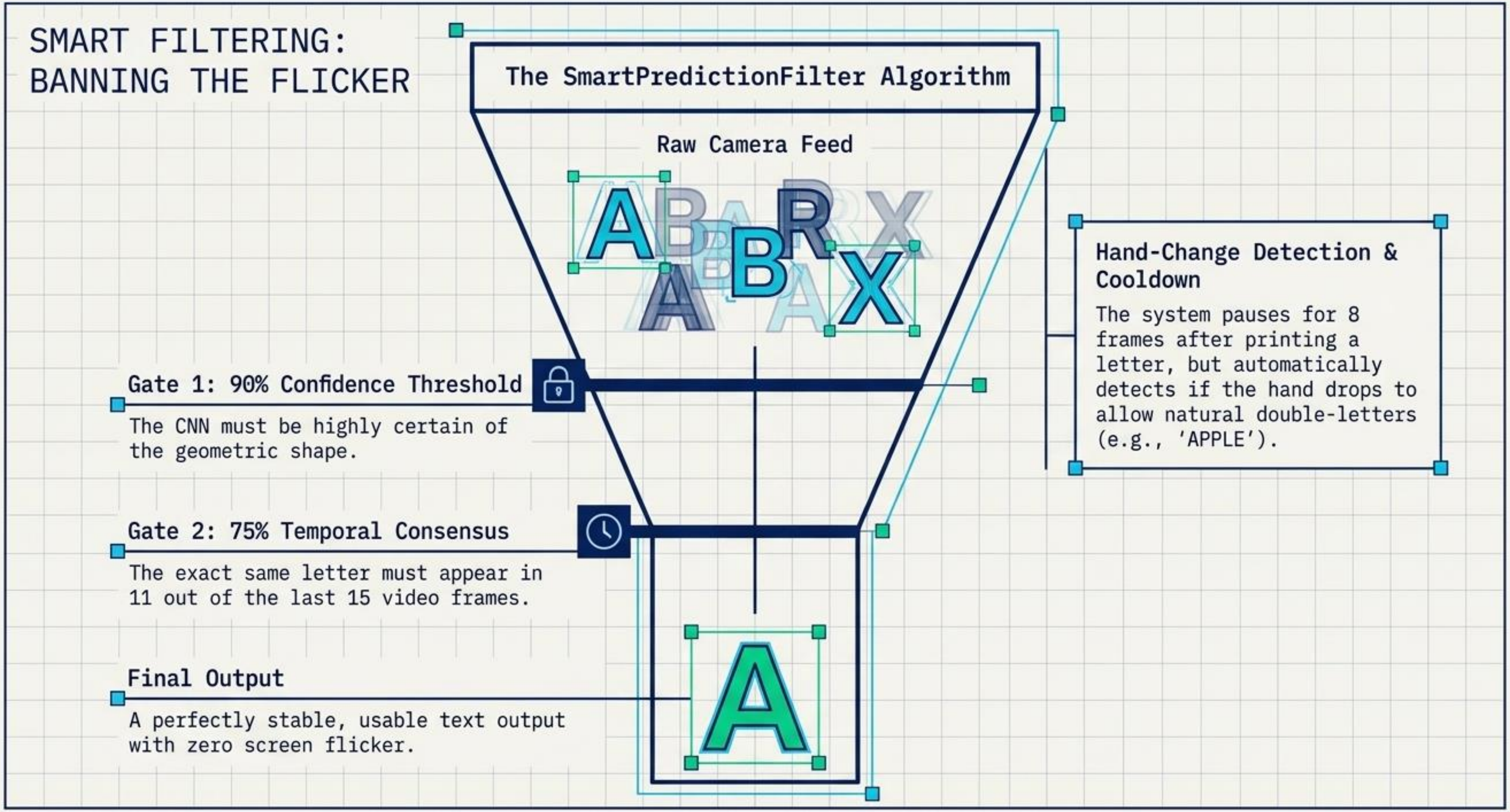
Dimensionality:
126 Features

The Zero-Padding Mechanism

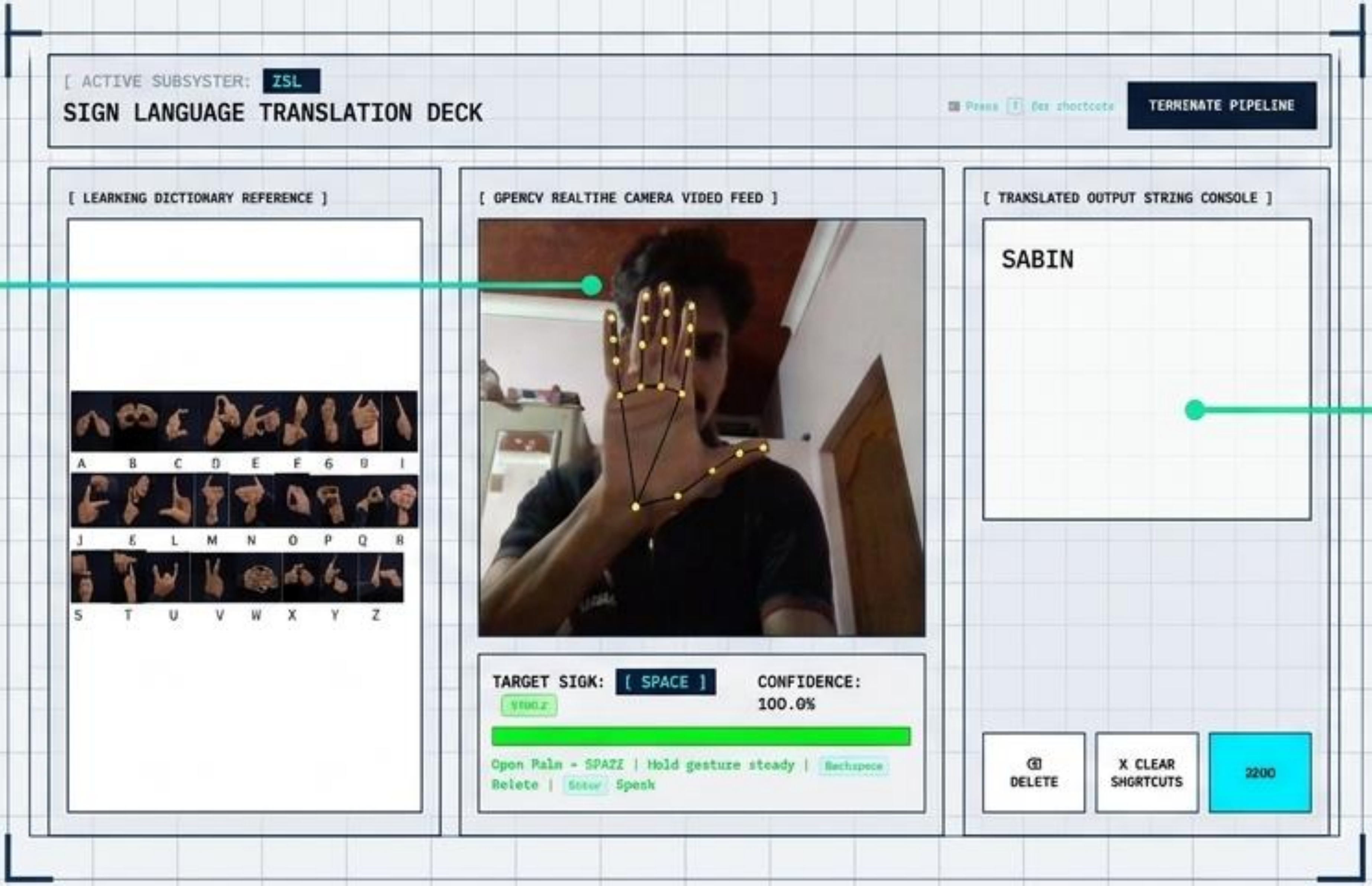


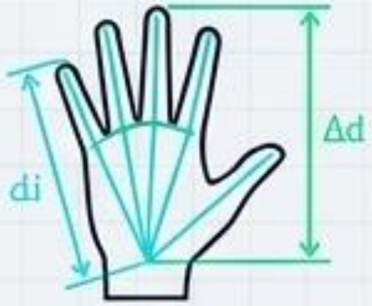

```
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…]
```

If an ISL sign only uses one hand, the system automatically zero-pads the missing 63 features to prevent network crashes, unifying both languages under one architecture.


SMART FILTERING:
BANNING THE FLICKER
The SmartPredictionFilter Algorithm
Raw Camera Feed
Gate 1: 90% Confidence Threshold
The CNN must be highly certain of the geometric shape.
Gate 2: 75% Temporal Consensus
The exact same letter must appear in 11 out of the last 15 video frames.
Final Output
A perfectly stable, usable text output with zero screen flicker.
Hand-Change Detection & Cooldown
The system pauses for 8 frames after printing a letter, but automatically detects if the hand drops to allow natural double-letters (e.g., 'APPLE').

THE USER EXPERIENCE





Hands-Free Spacing

Opening the palm triggers a SPACE. Uses pure deterministic geometry, requiring zero CNN compute power.

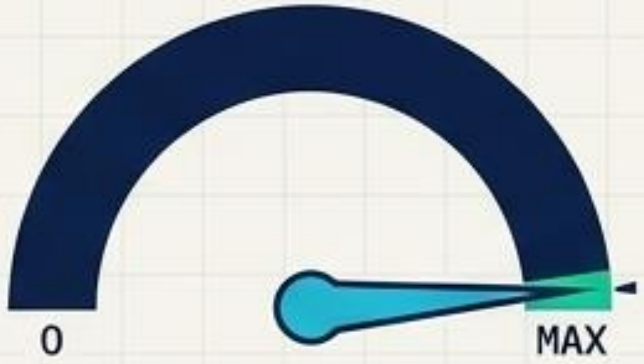
Offline Voice Synthesis

HTML5 Web Speech API automatically reads built sentences aloud. Operates entirely offline with zero internet requirement.

RESULTS & PERFORMANCE:

THE 99.9% MILESTONE

Accuracy



99.92% ASL

100% ISL

Proven via Confusion Matrices and per-class F1-scores.

Speed

< 5ms Inference

28-30 FPS

Sustained real-time performance on standard CPUs.

Weight

0.8 MB Model

Zero GPU Needed

Eliminates hardware dependencies entirely.

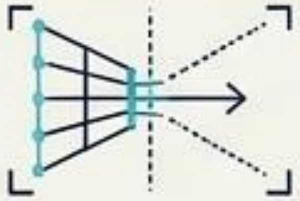
SIGN-BRIDGE proves that state-of-the-art accuracy and practical, real-world deployment on standard laptops are completely complementary goals.

CONCLUSION & **FUTURE IMPACT**

PROJECT IMPACT

The 15,000x Shift:

Proved that geometric landmarks are vastly superior to heavy pixel grids for accessible computing.

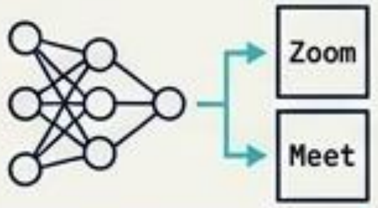


Unified Multi-Modal Setup:

Successfully bridged ASL and ISL into a single, offline browser application.

The Next Frontier:

Future integration with LSTMs for dynamic, moving signs and direct API hooks into Zoom/Meet.



DEFENSE & Q&A

Thank You to my guide Priyesh K. G., HOD Neetha P. C., and the evaluation panel.

Ready for Viva Q&A

- Lighting-Invariant Performance
- Flask Web Architecture
- Vanishing Gradient Solutions